

BrickAtlas Hierarchy: Crossed Evaluation of Brick Structure, Reasoning, and Executable Intervention

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Abstract

Brick reasoning involves both object complexity and task composition. Increasing the number of views or edits does not create independent objects, and naming a task “physical” does not validate its mechanics. We introduce the expanded Hierarchy-3 public development release of BrickAtlas, crossing 144 original digital layouts with 48 shared task families. Four structural strata contain 36 layouts each; every layout supports 22 atomic, 15 metacognitive/physical, six procedural, and five integrative tasks, for 6,912 questions on 42,617 visible parts. A state-transition interpreter checks submitted actions, while resource and possible-world validators evaluate scheduling and conditional policies. Rapier supplies actual collision queries and sampled dynamic traces for all objects. The original BrickAtlas website now exposes every layout as interactive 3D geometry, with exploded inspection and playback of submitted actions. A historical pilot on the retained 48-layout release yielded 7 successes on 16 fixed questions for GPT-4.1 mini at \$0.013621. This tests pipeline functionality, not calibrated difficulty or a model ranking. The revision corrects ambiguous choices, geometry envelopes, inter-module penetrations and replay state. All 144 layouts satisfy the declared nominal-state protocol, whose finite-force servos and conservative geometry remain explicit limitations.

1. Introduction

Digital brick environments offer discrete object identities, hierarchical assemblies, and explicit edits. These properties support reproducible evaluation across perception, structural reasoning, and intervention. However, benchmark growth can obscure rather than resolve a model’s failure: more derivatives of a few objects create larger denominators without broader source coverage. Likewise, a final correct object does not show that an agent constructed, repaired, or inspected it correctly.

BrickAtlas organizes evaluation along two axes. The vertical axis describes the structural scope of an object: component, assembly, mechanism, or system. The horizontal axis

describes task composition: atomic operations, metacognitive and physical reasoning, executable procedures, and integrated constraints. Crucially, the same 48 families occur at every level, so comparisons can hold task identity fixed. Figure 2 shows the complete crossed design.

This expansion retains all 48 corrected Hierarchy-3 layouts and their 2,304 tasks byte-for-byte, and adds 96 layouts with 4,608 tasks. Twelve construction families each contribute two configurations at four structural levels. These are 144 layouts, not 144 independently sampled mechanical families: new layouts reuse authored components. OMR/LDraw examples inform visual design but are not benchmark cases. The retired 5,120 procedural sources remain historical regression data and are excluded from the new scale.

Three implementation changes improve inspectability. First, all physics-derived questions use archived Rapier observations rather than handwritten response tables. Second, action answers execute a public state machine; an equivalent legal order can pass without matching a single reference list. Third, dataset geometry is available inside the original application, allowing inspection of modules, individual parts, exploded configurations, and intermediate action states.

We make the following contributions:

1. a balanced four-by-four organization with 144 authored layouts and 48 task families repeated on every layout;
2. executable action, schedule, and possible-world contracts with negative controls and explicit physical-evidence boundaries;
3. an integrated 3D browser sharing model data with the offline release and displaying reference or submitted programs; and
4. a versioned correction audit, semantic-answer baselines, and per-object geometry and nominal-state validation.

2. Related Work

BrickGPT [7] addresses stable brick generation; BrickNet [3] uses typed connection graphs. BrECS [1] studies sequential assembly with budgets. Hierarchy-3 evaluates

Level	Layouts	Parts/layout	Questions
D1	36	14–124	1728
D2	36	45–180	1728
D3	36	43–449	1728
D4	36	214–961	1728

Table 1. Actual release counts. D2 and D3 overlap in part count because the strata describe structural scope, not measured psychometric difficulty.

an existing design and its interventions; it does not claim comparable asset scale or a new generative method.

LEGO-Puzzles [8] and PhyBlock [5] cover visual, spatial, ordering, and counterfactual reasoning. Manual-based assembly methods such as MEPNet [10], Neural Assembler [11], and TreeSBA [2] recover poses or construction steps. Our contribution is a shared task-by-source organization with executable contracts, rather than a claim that each individual family is new.

Break and Make [9] evaluates interactive disassembly and reconstruction. WorkBenchMark [6] and BC-Bench [4] address embodied or compositional assembly. We explicitly separate digital state validity from real insertion force, connector tolerance, grasping, and hardware transfer.

3. Library and Crossed Design

3.1. Four Structural Strata

D1 contains local devices such as gimbals, grippers, hatches, and valve stands. D2 introduces complete assemblies including service lifts, loaders, trackers, and inspection craft. D3 contains articulated whole mechanisms such as cranes, observatories, robots, and excavators. D4 adds multi-workcell systems, including cargo terminals, docking yards, energy stations, and service hangars.

Table 1 reports actual counts rather than nominal target intervals. The library has 2,040 rigid modules and 1,938 declared joints. All layouts are agent-authored using reusable construction primitives; no independent human design study was conducted. Some new layouts share serial-stage constructors and differ in joint type, topology or branch count; layout count does not establish semantic independence. All 96 additions have distinct geometry fingerprints after removing names, colors and global translation. The retained indexed latch and orthogonal probe stage share geometry but differ in joint configuration; the complete collection has 143 distinct geometry fingerprints.

New constructions include arms, grippers, carousels, cross-feed stages, cradles, drums, gantries, folding panels, elastic suspensions, gates, inclined probes and turntables. D2 adds a movable carrier and console, D3 combines two stations with storage, and D4 combines three or four stations with shared inspection. This creates matched construc-

tion families across levels, without claiming independent design samples.

3.2. Representation and Provenance

A part has a shape, size, local position, quaternion, color, and module membership. Modules have world poses, masses, roles, and anchoring status. Joints identify endpoints, local anchors, type, axis, and limits. The visible vocabulary includes bricks, plates, beams, windows, slopes, arches, cylinders, wheels, axles, gears, and spheres.

These shapes are LEGO-like digital primitives, not verified commercial part specifications. Original-site model JSON and offline model JSON are byte-identical. Each object also records its family and lineage; future train/test splits must group related layouts and reused components. The present public development set has no hidden confirmation split.

3.3. Matched Coverage

Every level has 1,728 questions. Atomic tasks are repeated across levels, including recognition, counting, local edits, topology, and pose corrections. Meta tasks include both physical-record interpretation and uncertainty-aware decisions, with new nonuniform posteriors, expected-loss decisions and force-moment calculations. Procedural families include assembly, disassembly, supported service repair, compound editing, guarded repair and rollback. Integration adds exclusive-tool repair and budget-constrained policies to distributed repair, scheduling and inspection-conditioned policies.

Task identifiers bind source and family. Changing a render, color, answer order, or fault creates a condition within a source, not a new source object. Although all families are crossed, many local questions remain easy with explicit structured input; a larger scene alone does not guarantee a harder local operation.

4. Task and Scoring Contracts

4.1. Atomic and Metacognitive Questions

Atomic tasks return a choice ID. Each distractor changes a relevant field, set, count, identity, or transformation. Option rotation is deterministic per task. Set-valued answers are canonicalized before deduplication; ordered vectors and programs retain their ordering. Some questions have fewer than four distinct alternatives; chance baselines must use the actual option count.

Multi-answer tasks require the entire feasible set. In accessibility, an empty feasible set is valid if every proposed path collides. Counterfactual support excludes the explicitly removed object and recomputes reachability to any anchored root through remaining joint edges, allowing redundant support paths.



Figure 1. **Representative layouts at all four structural levels.** Rows progress from D1 to D4; three fixed representatives per level are shown. These are canvas renders from the original application, not generated concept images. All 144 layouts and all four views per layout are provided in the supplementary index.

Inspection uses a finite uniform prior and deterministic observations. For query q with observation partition sizes n_o over N worlds,

$$U(q) = \frac{-\sum_o (n_o/N) \log_2(n_o/N)}{c(q)}. \quad (1)$$

All utility-maximizing queries are accepted. Abstention tests whether all compatible worlds imply the same action. Posterior tasks condition the finite prior on an observed query result. Pareto tasks return every non-dominated alternative under cost, mass, and stiffness. Additional tasks use explicit nonuniform prior weights and asymmetric failure losses. Six previously constant-answer families now vary their actual parameters and semantic answers; the release reports per-family semantic-majority baselines, not just letter-choice baselines.

Public inputs explicitly disclose structural metadata

where required. Therefore most atomic scores measure structured reasoning with auxiliary images, not isolated visual perception. Image-removal, symbolic-removal, and viewpoint ablations are necessary before claiming visual reasoning gains.

4.2. Executable Procedures

An action is a tuple

$$a = (P_a, F_a, A_a, D_a, c_a), \quad (2)$$

with required facts, forbidden facts, add effects, delete effects, and cost. From state S , execution requires $P_a \subseteq S$ and $F_a \cap S = \emptyset$, then applies

$$S' = (S \setminus D_a) \cup A_a. \quad (3)$$

The interpreter checks every action in submitted order, stops at the first invalid transition, and records its cause. Success

Hierarchy-3 expanded: 144 layouts $\times$ 48 task families

	Atomic	Meta / physics	Execution	Integration
D1 Components	792 questions 36 sources / level	540 questions 36 sources / level	216 questions 36 sources / level	180 questions 36 sources / level
D2 Assemblies	792 questions 36 sources / level	540 questions 36 sources / level	216 questions 36 sources / level	180 questions 36 sources / level
D3 Mechanisms	792 questions 36 sources / level	540 questions 36 sources / level	216 questions 36 sources / level	180 questions 36 sources / level
D4 Systems	792 questions 36 sources / level	540 questions 36 sources / level	216 questions 36 sources / level	180 questions 36 sources / level

Coverage counts, not measured model accuracy.

Figure 2. **Crossed structural scope and task composition.** Each cell contains 36 sources. Every source contributes 22 atomic, 15 meta/physical, six procedural, and five integrative questions. The axes are design strata; measured monotonic difficulty is not assumed.

requires all goal facts and the declared budget.

Assembly places a module only after its declared predecessors. Disassembly removes children before parents. Repair sequences support, open, remove, replace, verify, close, and release an interface. Compound editing substitutes a color operation within a protected service sequence. Independent actions can commute. Duplicate placements, unknown IDs, missing terminal steps, or budget overruns fail. Guarded repair and rollback expose unsafe shortcuts that cannot satisfy verified goals. Exclusive-tool tasks remove a shared availability fact until the active intervention releases the tool.

The transition contract certifies symbolic intervention rules. It does not automatically simulate robot motion or tool contact for every repair action. The website changes module visibility and colors according to these same transitions and displays state facts alongside the 3D replay, so presentation and scoring share one interpreter.

4.3. Integrated Constraints

Distributed repair combines multiple independent faulty modules under one budget and one final goal. Scheduling requires integer start times, predecessor completion, non-overlapping use of exclusive workcells, and completion by a deadline. Inspection policies select a query and map every possible observation to an action; the validator checks

all worlds instead of only a sampled rollout.

No LLM judge determines correctness. Empty or malformed outputs remain failures. A batch evaluator must retain missing cases in its frozen denominator and reject unknown or duplicate case IDs. Per-family and per-source reporting is preferred to a pooled percentage.

5. Physics and Quality Audit

Rapier 0.20.0 runs at 120 Hz with 64 solver iterations. Visible parts are conservatively approximated by stud-inclusive cuboids and grouped into rigid bodies. Fixed, revolute, prismatic, and spring descriptors instantiate the declared mechanism. Bases are anchored; holding motors use public force-based stiffness 60,000, damping 1,500 and maximum force 20,000 in scene units. A separate static world performs three continuous service shape casts per object, excluding the serviced module.

After 120 nominal steps, the audit records a geometric representative point’s displacement, angular displacement, and non-spring joint residuals. All 144 objects satisfy fixed thresholds of 0.35 scene units, 0.15 radians, and 0.02 scene units respectively. Every rendered vertex lies within its envelope. All joint anchor pairs initially coincide, and oriented cuboid contact queries find no penetration above 0.005 between any module pair, including directly joined modules. Joint contacts remain enabled. Revolute limits

GPT-4.1 mini · 16-case pipeline pilot

	Atomic	Meta / physics	Execution	Integration
D1	1/1 n=1 (pilot)	0/1 n=1 (pilot)	1/1 n=1 (pilot)	0/1 n=1 (pilot)
D2	0/1 n=1 (pilot)	0/1 n=1 (pilot)	1/1 n=1 (pilot)	0/1 n=1 (pilot)
D3	1/1 n=1 (pilot)	0/1 n=1 (pilot)	0/1 n=1 (pilot)	0/1 n=1 (pilot)
D4	1/1 n=1 (pilot)	0/1 n=1 (pilot)	1/1 n=1 (pilot)	1/1 n=1 (pilot)

Descriptive cells only. Not population accuracy or calibrated difficulty.

Figure 3. Historical pilot on retained base-48 objects. A green cell means one correct response, not established 100% population accuracy.

are set and read back on instantiated joints, with a driven over-limit regression control.

Corrections relocate interfering service modules, ruders, drills, racks, pumps and transfer stages, repair four inconsistent tower anchors, and add separated elastic moorings to the flood-response barge. These corrections and the change from 120/20/100 motor settings were introduced in the retained 48-layout release. The expansion uses that same protocol and thresholds. Active finite-force servos are not passive stability or real LEGO material calibration.

Impulse application and response measurement use the geometric representative point, remaining invariant under a change of local origin. An off-center impulse is followed by 120 simulation steps, sampled at eleven time points. Dynamic questions ask for the maximum of the *sampled translational response*, not the continuous maximum, a rotational safety bound, or a calibrated load threshold. They cannot be used to claim real-world robustness.

Other checks cover unique identities, finite poses, positive dimensions, declared dependency order, rendered-image hashes, and exact website/ dataset parity. Inter-module bounding-box overlaps are published as diagnostics rather than silently treated as precise mesh collisions.

6. Pilot and Inspectable Release

The historical fixed pilot selected local-frame transformation, expected loss, guarded repair, and resource repair at each level. GPT-4.1 mini receives one image and public structured evidence at temperature zero, with a 2,200-token output cap, no automatic retries, and no provider fallback. The 16 responses yield 7 exact successes at \$0.013621; raw replies, provider IDs, costs, and public-input hashes are archived. Table 2 preserves each denominator. No new paid model calls were made for the 96 additions; these historical

Level	Atomic	Meta	Procedure	Integrated
D1	1/1	0/1	1/1	0/1
D2	0/1	0/1	1/1	0/1
D3	1/1	0/1	0/1	0/1
D4	1/1	0/1	1/1	1/1

Table 2. GPT-4.1 mini pipeline pilot. Each cell is one fixed question. These are observed counts, not estimates of whole-cell accuracy.

counts are not accuracy measurements on the 144-layout expansion.

Deterministic checks accept all 6,912 reference answers. A separate public-precondition solver succeeds on 1,152 action tasks. The simple constant-choice/empty-answer control succeeds on 1340/6,912 cases, with heterogeneous choice cardinalities and output contracts. Reference acceptance is evaluator validation, not learned-model performance; the action solver has privileged symbolic rules compared with a purely visual agent.

The original application’s homepage lists all 144 benchmark models. Each opens an orbit-controlled 3D workspace with zoom, pan, canonical views, module isolation, explosion, and image export. Its task panel shows all 48 contracts, and action JSON can be replayed with a step slider and first-failure feedback. Selection and isolation preserve the current frame’s active modules. The release contains 192 retained views and 384 new captures, totaling 576 audited model views. The human review dashboard reads its denominator from the catalog; existing task IDs and feedback remain compatible. Every source and every task family is indexed in the supplement; full question/choice/input/answer records are shipped in the Chinese question bank and machine-readable release.

7. Limitations and Next Validation

The release substantially increases coverage and inspectability, but the structural levels remain authored categories. There is no human timing study, item-response calibration, or causal proof that higher levels are harder. Shared constructors create correlated layouts. The 16-question pilot is too small for population accuracy, robust ranking, or level-wise discrimination claims.

Physical realism is limited by compound bodies, cuboid envelopes, anchored bases, and uncalibrated material parameters. Digital primitive connectivity is not commercial LEGO clutch-force validity. Passing the declared nominal-state protocol does not certify arbitrary motion, passive balance, or extreme loads. Symbolic repair transitions and exploded display layouts are not collision-free robot trajectories.

Future confirmation requires source-family-disjoint hid-

den designs, manual geometry review, calibrated connectors and contact mechanics, human difficulty measurements, and information-condition ablations. These are outstanding scientific requirements, not outcomes inferred from passing software tests.

8. Conclusion

Hierarchy-3 provides a common source-by-task organization for brick reasoning, execution, and integrated decisions. Its 144 layouts, 6,912 questions, state-based scoring, actual simulator records, and original-site 3D access make the benchmark easier to inspect and extend. Versioned errata preserve historical inputs and scores while the corrected release remains explicit about physical and statistical boundaries.

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